

=====

```
select employee_id,  
employee_name,  
department,  
salary,  
hire_date,  
ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary) as  
row_num  
from employees;
```

ROW_NUMBER
RANK
DENSE_RANK

<WINDOW FUNCTION> OVER (PARTITION BY <partition column> ORDER BY <order by col>)

ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary desc)

WHAT IS THE ROLE OF GROUP BY?

to aggregate and for each group we get a single row

but using window functions the input and output
has the same number of rows for each partition

-- we can remove the partition by clause also

```
select employee_id,  
employee_name,  
department,  
salary,  
hire_date,  
ROW_NUMBER() OVER (ORDER BY salary desc) as row_num  
from employees;
```

-- can we remove the order by also?

```
select employee_id,  
employee_name,  
department,  
salary,  
hire_date,  
ROW_NUMBER() OVER () as row_num  
from employees;
```

-- partition by is optional but order by is mandatory

```
CREATE TABLE employees (  
  employee_id INT PRIMARY KEY,  
  employee_name VARCHAR(50),  
  department VARCHAR(50),  
  city VARCHAR(50),  
  salary DECIMAL(10, 2),  
  hire_date DATE  
);
```

```
INSERT INTO employees (employee_id, employee_name, department, city, salary, hire_date)  
VALUES  
(1, 'Amit', 'HR', 'Mumbai', 50000, '2022-01-15'),  
(2, 'Neha', 'HR', 'Mumbai', 55000, '2023-03-10'),  
(3, 'Suresh', 'HR', 'Delhi', 48000, '2021-11-20'),  
(4, 'Rohit', 'HR', 'Delhi', 52000, '2022-09-05'),  
(5, 'Raj', 'Finance', 'Mumbai', 60000, '2021-07-23'),  
(6, 'Ravi', 'Finance', 'Delhi', 62000, '2022-09-01'),  
(7, 'Kiran', 'Finance', 'Mumbai', 58000, '2021-02-14'),  
(8, 'Sunita', 'Finance', 'Delhi', 61000, '2023-01-11'),  
(9, 'Priya', 'IT', 'Mumbai', 70000, '2020-12-02'),  
(10, 'Anjali', 'IT', 'Delhi', 67000, '2021-11-19'),  
(11, 'Vikas', 'IT', 'Mumbai', 69000, '2022-05-20'),  
(12, 'Sanjay', 'IT', 'Delhi', 72000, '2023-04-30'),  
(13, 'Meena', 'IT', 'Delhi', 68000, '2021-03-15');
```

```
select employee_id,  
employee_name,  
department,  
city,  
salary,  
hire_date,  
ROW_NUMBER() OVER (PARTITION BY department,city ORDER BY salary  
desc) as row_num  
from employees;
```

-- inserts with few duplicate salaries

```
INSERT INTO employees (employee_id, employee_name, department, city, salary, hire_date)  
VALUES  
(1, 'Amit', 'HR', 'Mumbai', 50000, '2022-01-15'),  
(2, 'Neha', 'HR', 'Mumbai', 55000, '2023-03-10'),  
(3, 'Suresh', 'HR', 'Delhi', 48000, '2021-11-20'),  
(4, 'Rohit', 'HR', 'Delhi', 52000, '2022-09-05'),  
(5, 'Raj', 'Finance', 'Mumbai', 60000, '2021-07-23'),  
(6, 'Ravi', 'Finance', 'Delhi', 60000, '2022-09-01'),  
(7, 'Kiran', 'Finance', 'Mumbai', 58000, '2021-02-14'),
```

```
(8, 'Sunita', 'Finance', 'Delhi', 60000, '2023-01-11'),  
(9, 'Priya', 'IT', 'Mumbai', 70000, '2020-12-02'),  
(10, 'Anjali', 'IT', 'Delhi', 68000, '2021-11-19'),  
(11, 'Vikas', 'IT', 'Mumbai', 68000, '2022-05-20'),  
(12, 'Sanjay', 'IT', 'Delhi', 72000, '2023-04-30'),  
(13, 'Meena', 'IT', 'Delhi', 68000, '2021-03-15');
```

```
select employee_id,  
employee_name,  
department,  
city,  
salary,  
hire_date,  
ROW_NUMBER() OVER (PARTITION BY department,city ORDER BY salary desc, hire_date) as  
row_num  
from employees;
```

```
select employee_id,  
employee_name,  
department,  
city,  
salary,  
hire_date,  
ROW_NUMBER() OVER (ORDER BY salary desc, hire_date) as row_num  
from employees;
```

-- the below is not going to work

```
select employee_id,  
employee_name,  
department,  
city,  
salary,  
hire_date,  
ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary desc, hire_date) as row_num  
from employees  
where row_num <=2;
```

-- using subquery we can achieve it

```
select * from (select employee_id,  
employee_name,  
department,  
city,  
salary,  
hire_date,  
ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary desc, hire_date) as row_num  
from employees) e where row_num <=2;
```

-- convert the above to CTE

```
with emp_cte as (select employee_id,  
employee_name,  
department,  
city,  
salary,  
hire_date,  
ROW_NUMBER() OVER (PARTITION BY department ORDER BY salary desc, hire_date) as row_num  
from employees)  
select * from emp_cte where row_num <=2;
```

group by vs window function